THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 3

Chapter 4: **Electricity: Magnetic and Heating Effects** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which of these changes electrical energy to heat? 1

(a) electric fan **(b)** cell **(c)** loudspeaker **(d)** electric iron

2. An electromagnet is made by winding a coil of insulated wire around a: 1

(a) wooden rod **(b)** copper rod **(c)** soft iron core **(d)** plastic rod

3. Wires carrying current are covered with plastic because: 1

(a) plastic produces heat **(b)** plastic is cheaper than gold
(c) plastic is a conductor **(d)** plastic is an insulator

4. A battery is: 1

(a) a bulb **(b)** a switch
(c) a conductor **(d)** a combination of two or more cells

5. An electromagnet loses its magnetism when: 1

(a) it is large **(b)** the core is iron
(c) the current is switched off **(d)** the current is on

6. The unit of electric current is the: 1

(a) ampere **(b)** volt **(c)** joule **(d)** ohm

7. Electromagnets are used in: 1

(a) cloth **(b)** plastic toys
(c) candles **(d)** cranes that lift scrap iron

8. The heating effect of electric current is used in: 1

(a) a compass **(b)** an electric heater
(c) a magnet **(d)** a cell

9. The cell symbol has: 1

(a) a zigzag **(b)** two equal lines
(c) a circle **(d)** a long line and a short thick line

10. Which is a safe practice? 1

(a) overloading sockets **(b)** touching a bare wire
(c) using wet hands on a switch **(d)** switching off appliances before repair

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. What is meant by overloading? 2

12. What is an electromagnet? 2

13. Name two devices that work on the magnetic effect of current. 2

14. Why should electrical appliances be handled with dry hands? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Differentiate between a permanent magnet and an electromagnet. 3

16. Give three differences between a fuse and an MCB. 3

17. Why do lamps connected in a house get the full voltage instead of sharing it? 3

OR

Explain the working of an electric bell in brief. 3

18. List three uses of electromagnets. 3

OR

State three reasons why a bulb may not glow even when connected to a cell. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. What is a fuse and an MCB? Explain how they protect appliances and mention the causes of excessive current. 5

OR

Explain the circuit symbols of a cell, switch (open and closed), bulb, resistor and wire, and draw their meaning in words. 5

20. Explain the difference between conductors and insulators using a circuit activity with a bulb. Give three examples of each. 5

OR

Explain why short circuits and overloading are dangerous and how they are prevented. 5

END OF QUESTION PAPER — Total: 40 Marks

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